

- Hide menu
- Week 2: Image Formation
- Reading: Week 2 Lecture Handout

30 min
- Ungraded Plugin: 2.1

Overview of Image Formation

3 min
- Practice Quiz: 2.1

Overview of Image Formation Self-check Quiz

Started
- Ungraded Plugin: 2.2

Pinhole and Perspective Projection

21 min
- Practice Quiz: 2.2

Pinhole and Perspective Projection Self-check Quiz

15 min
- Reading: 2.3

Video Correction

10 min
- Ungraded Plugin: 2.3

Image Formation using Lenses

12 min
- Practice Quiz: 2.3

Image Formation using Lenses Self-check Quiz

15 min
- Ungraded Plugin: 2.4

Depth of Field

15 min
- Practice Quiz: 2.4

Depth of Field Self-check Quiz

15 min
- Ungraded Plugin: 2.5

Lens Related Issues

8 min
- Practice Quiz: 2.5

Lens Related Issues Self-check Quiz

5 min
- Ungraded Plugin: 2.6

Wide Angle Cameras

24 min
- Ungraded Plugin: 2.7

Animal Eyes

16 min
- Discussion Prompt: Week 2

Intrusive Detection System

10 min
- Week 2 Quiz

✔ Congratulations! You passed!

Grade received 100%

To pass 50% or higher

Go to next item

2.2 Pinhole and Perspective Projection Self-check Quiz

1. Consider that we are working with a pinhole camera and an object is present at a distance of z_o in front of the camera. How would the magnification be affected if we moved the object away from the camera?

1 / 1 point

coach

Ready to review what you've learned before taking the quiz? I'm here to help.

✦ Help me practice

✦ Let's chat

☒ The magnification would decrease

☐ The magnification would increase

☐ The magnification would remain the same

☐ None of the above

✔ Correct

Magnification is inversely proportional to the distance of objects in the real world. Therefore, if we move the object away from the camera, the magnification decreases.

✔ Receive grade

To Pass 50% or higher

Your grade 100%

View Feedback

We keep your highest score

1 / 1 point

Try again

2. Consider two blurry images captured from two cameras A and B. A has diameter less than ideal size and B has diameter more than ideal size. Is it possible to tell which camera captured which image?

1 / 1 point

☐ Yes

☒ No

☐ Yes, if the diameter and focal length are given.

✔ Correct

Due to phenomena such as diffraction if the diameter of the pinhole is less than ideal diameter then the light will diffract and produce a blurry image similar to the ones produced by camera with diameter larger than ideal diameter. Therefore, it is not possible to identify the source of blur.

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3. Given a pinhole camera whose focal length is 2mm. Calculate the coordinates of the vanishing point of all the parallel lines in the real world which are parallel to the line passing through the point (0.1m, 0.3m, 0.1m) and the pinhole.

1 / 1 point

☒ 2mm, 6mm

☐ 2m, 6m

☐ 1mm, 0.66mm

☐ 1m, 0.66m

✔ Correct

Use the following formula: $(x, y) = (f * l_x / l_z, f * l_y / l_z)$

https://www.coursera.org/learn/cameraandimaging/quiz/50aiv/2.2-pinhole-and-perspective-projection-self-check-quiz?attemptRedirectToCover=true

1/1